

Botond Branyicskai-Nagy

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Education

University College London	September 2023 – (September 2024)
MSc Machine Learning	
Probabilistic and Unsupervised Learning ¹ Approximate Inference ¹ Applied ML	
Bayesian Deep Learning Supervised Learning Reinforcement Learning ²	
Applied Deep Learning ML Seminar	
Imperial College London	2020 – 2023
BSc Physics with Theoretical Physics	
Eotvos Jozsef Secondary School, Budapest	2014 – 2020
A-levels equivalent in Hungary at advanced level:	
Mathematics, Physics, English	

Research Experience

Master's Thesis - University College London	2024
Learning mathematical abstraction: investigating properties of learnt representations and drawing conclusions as to how to better guide learning in a neural network for downstream tasks like mathematical reasoning. Supervised by Prof. Mirco Musolesi.	
Research Internship — Imperial College London	2023
Causal discovery using cross-validation as compared with Bayesian Model Selection (GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP.	
Computational Evolution — Imperial College London	2023
BSc project: Simulation and analysis of evolution by tracking species of agents in varying environments. Comparison with theoretical models, exploration of optimisation algorithms.	

Other Projects

Barabasi-Albert and Oslo Models — Imperial College London	2023
Implementation of a simulation and analysis of each model's behaviour.	
Solar Sail (Project Svarog) — Imperial College Space Society	2022
Orbital mechanics: spacecraft trajectory simulation, atmospheric effects, relativistic corrections.	
Machine learning for LHCb experiment — Imperial College London	2022
Leading ML team in project analysing decay product distribution from LHCb experiment for Standard Model comparison. Neural Networks (TensorFlow), Decision Trees.	
Laser microphone — Imperial College London	2020
Summer project (group of 4): building a proof-of-concept prototype of a laser interferometer-based device capable of recovering audio from vibrations of a window from a distance.	

Awards

2nd Year Quantum Physics Article Prize – Published on Imperial Website (2022)
Young Researchers of Natural Sciences, Conference Presentation Distinction (2016)

Skills and Interests

Languages & Programming
Python - Scientific and ML libraries (TensorFlow, GPflow, PyTorch), Julia, Cluster usage
Languages
Hungarian (Native), English (Proficient - Cambridge Advanced 206), French (Intermediate)
Other Interests: Space Society, tennis, hiking, piano

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind

For academic references please contact:

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Professor Daniel Waldram

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